

SL-1, W02-L01

Learning paradigms and regression

Transfer learning: applying the knowledge obtained from one problem or domain to solve a new but related problem instead of starting from scratch.

Reinforcement learning: It is a learning process through trial and error, with rewards for good decisions and penalties for bad decisions.

Supervised learning:

- Regression
- Classification

Regression

Model:

$$Y = f(x) + \varepsilon \text{response} = \text{signal} + \text{noise}$$

In regression, our aim is to estimate the signal/regression function f .

In order to estimate f , we need to consider a loss function and minimize the expected loss.

Popular loss functions:

- Squared error loss

$$L(Y, f(x)) = (Y - f(x))^2$$

- Absolute error loss

$$L(Y, f(x)) = |Y - f(x)|$$

We should try to find that minimizes Risk $= R(f) = E(L(Y, f(x)))$, the expected loss. In practice, we minimize the empirical analog of $E(L(Y, f(x)))$. This is roughly

$$\frac{1}{n} \sum L(Y_i, f(x_i)),$$

where $(X_1, Y_1), \dots, (X_n, Y_n)$ are independent observations on (X, Y) .

For example, in the case of squared error loss, we minimize

$$\frac{1}{n} \sum (Y_i - f(x_i))^2 = \sum (Y_i - f(x_i))^2.$$

Similarly, for absolute error loss function, we minimize

$$\sum |Y_i - f(x_i)|.$$

Bias-variance tradeoff and parametric regression

Model $Y = X + \varepsilon$ where $\varepsilon(t) \sim (0, \sigma^2)$

$$f(x) = ax + b \longrightarrow \text{underfit } \{\text{high bias, low var}\}$$

$$f(x) = ax^2 + bx + c \longrightarrow \text{fit}$$

$$f(x) = a_0x^0 + a_1x^1 + \dots + a_kx^k \longrightarrow \text{overfit } \{\text{high var, low bias}\}$$

$$E(Y - \hat{f}(x))^2 = E(Y - E(\hat{f}(x)))^2 + E(\hat{f}(x) - E(\hat{f}(x)))^2 = \text{bias}^2 + \text{variance}$$

f can be estimated either parametrically or nonparametrically.

In the case of parametric regression we assume a finite dimensional parametric form for f and estimate this parameters from the data.

For instance, in the case of linear regression, we assume f to be linear, i.e.

$$f(x) = \alpha + \beta x$$

and estimate this parameters by min

$$\sum (Y_i - \alpha - \beta x_i)^2$$

or

$$\sum |Y_i - \alpha - \beta x_i|.$$

We can think of even a simpler model $Y = \beta x$, where one minimizes

$$\sum (Y_i - \beta x_i)^2 \quad \text{or} \quad \sum |Y_i - \beta x_i|.$$

Simple model, LS weights, and LAD

$$\beta_1 = \frac{Y_1}{X_1}$$

$$\beta_2 = \frac{Y_2}{X_2}$$

$$\vdots$$

$$\beta_n = \frac{Y_n}{X_n}$$

$$\hat{\beta}_{LS} = \frac{\sum X_i Y_i}{\sum X_i^2} = \frac{\sum X_i^2 \beta_i}{\sum X_i^2} = \frac{\sum w_i \beta_i}{\sum w_i} \quad \text{where } w_i \propto X_i^2$$

$\hat{\beta}_{LAD}$ is the median of the following frequency distribution:

- Values $\beta_1, \beta_2, \dots, \beta_n$
- Weights $|X_1|, |X_2|, \dots, |X_n|$

$$|X_i| > \frac{1}{2} \sum |X_j|$$

$$\beta_1, \beta_2, \dots, \beta_n \quad |X_1|, \dots$$

Example:

$$X : 1, 2, 4, 8, \dots, 2^k$$

$$Y : 1, 2, 3, 4, \dots, k+1$$

$$\beta = Y/X : 1/1, 2/2, 3/4, 4/8, \dots, (k+1)/2^k$$

$$\beta_1, \beta_2, \beta_3, \dots, \beta_{k+1}$$

$$\min \sum |Y_i - \beta X_i|$$

$$\hat{\beta}_{LAD} = \frac{k+1}{2^k}$$

Another example:

$$X : 3, 4, 5, 7, 9$$

$$Y : 5, 10, 10, 30, 5$$

$$\text{Sum: } 6, 15, 25, 55, 60$$

For the general case, i.e. for the model

$$Y = \alpha + \beta X + \varepsilon$$

we have

$$\hat{\beta}_{LS} = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

$$\hat{\alpha}_{LS} = \bar{Y} - \hat{\beta}_{LS} \bar{X}$$

$(\hat{\alpha}_{LAD}, \hat{\beta}_{LAD})$ does not have a closed form expression.

$\sum |Y_i - \alpha - \beta X_i|$ is minimized using an iterative algo (namely the IRWLS algo is used).

Properties of LS solutions

Consider a model:

$$Y_i = \alpha + \beta X_i + \varepsilon_i,$$

where X_i 's are non-stochastic and the ε_i 's are iid with mean 0 and var σ^2 .

$$\begin{aligned} E(\hat{\alpha}) &= \alpha & | & & Var(\hat{\alpha}) &=? \\ E(\hat{\beta}) &= \beta & | & & Var(\hat{\beta}) &=? \\ Cov(\hat{\alpha}, \hat{\beta}) &=? \end{aligned}$$

In this more general case we consider a linear model of the form

$$\begin{aligned} Y_1 &= \beta_0 + \beta_1 X_{11} + \cdots + \beta_k X_{k1} + \varepsilon_1 \\ &\vdots \\ Y_n &= \beta_0 + \beta_1 X_{n1} + \cdots + \beta_k X_{nk} + \varepsilon_n \end{aligned}$$

$\varepsilon_1, \dots, \varepsilon_n$ are iid with mean 0 and var σ^2

We write it as

$$Y = X\beta + \varepsilon,$$

where

$$Y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \quad X = \begin{bmatrix} 1 & x_{11} & \cdots & x_{k1} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{kn} \end{bmatrix}, \quad \beta = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_k \end{pmatrix}, \quad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

In this case, we have

$$\begin{aligned} \hat{\beta} &= (X^T X)^{-1} X^T Y \\ E(\hat{\beta}) &= \beta \\ Var(\hat{\beta}) &= \sigma^2 (X^T X)^{-1} \end{aligned}$$

$$E\|\hat{\beta} - \beta\|^2 = \sigma^2 \text{trace}[(X'X)^{-1}] = \sigma^2 \sum_{i=1}^{k+1} \frac{1}{\lambda_i}$$

where λ_i 's are the eigenvalues of $X^T X$.

$$\|\hat{\beta} - \beta\|^2 = (\hat{\beta}_0 - \beta_0)^2 + \dots + (\hat{\beta}_k - \beta_k)^2 = (\hat{\beta} - E(\hat{\beta}))^2$$

In the case of multicollinearity, i.e. when $X^T X$ is near singular, $E(\hat{\beta} - \beta)^T(\hat{\beta} - \beta)$ will be high, so the least square estimates will not be reliable.

A simpler multicollinearity example

Let us consider a simpler example with the model

$$Y_i = \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i;$$

where the ε_i 's are iid with mean 0 and var σ^2 .

Also assume that all columns are standardized.

$$\begin{array}{ccc} Y_i & X_{1i} & X_{2i} \\ Y_i & X_{1i} & X_{2i} \\ \vdots & \vdots & \vdots \\ Y_n & X_{1n} & X_{2n} \end{array} \quad \begin{array}{l} \bar{Y} = 0 = \bar{X}_1 = \bar{X}_2 \\ S_Y^2 = 1 = S_1^2 = S_2^2 \end{array}$$

$$\hat{\beta} = \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = (X'X)^{-1}X'Y$$

$$V(\hat{\beta}) = \sigma^2(X'X)^{-1}$$

$$\hat{\beta} = \frac{1}{1 - \rho^2} \begin{bmatrix} 1 & -\rho \\ -\rho & 1 \end{bmatrix} \times \begin{bmatrix} \rho Y_1 \\ \rho Y_2 \end{bmatrix}$$

$$= \frac{1}{1 - \rho^2} \begin{Bmatrix} \rho Y_1 - \rho^2 Y_2 \\ \rho Y_2 - \rho^2 Y_1 \end{Bmatrix}$$

$$V(\hat{\beta}) = \sigma^2(X'X)^{-1}$$

$$= \frac{\sigma^2}{n(1 - \rho^2)} \begin{bmatrix} 1 - \rho & -\rho \\ -\rho & 1 \end{bmatrix}$$

$$\text{Var}(\hat{\beta}_1) = \text{Var}(\hat{\beta}_2) = \frac{\sigma^2}{n(1 - \rho)}$$

Note:

$$E(\hat{\beta} - \beta)'(\hat{\beta} - \beta) = \sum_{i=0}^k Var(\hat{\beta}_i)$$

$$Var(1'\hat{\beta}) = 1'\sigma^2(X'X)^{-1}1$$

Now,

$$X'X = \begin{bmatrix} \sum X_{1i}^2 & \sum X_{1i}X_{2i} \\ \sum X_{1i}X_{2i} & \sum X_{2i}^2 \end{bmatrix}$$

$$X = \begin{bmatrix} X_{11} & X_{21} \\ X_{12} & X_{22} \\ X_{13} & X_{23} \\ \vdots & \vdots \\ X_{1n} & X_{2n} \end{bmatrix}$$

$$= \begin{bmatrix} n & n\rho \\ n\rho & n \end{bmatrix}$$

$$= n \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix} \quad \text{where } \rho = \text{corr}(X_1, X_2).$$

$$X'Y = \begin{bmatrix} \sum X_{1i}Y_i \\ \sum X_{2i}Y_i \end{bmatrix} = n \begin{bmatrix} \rho Y_1 \\ \rho Y_2 \end{bmatrix}, \quad \text{where } \rho_{Y_i} = \text{corr}(Y, X_i), \quad i = 1, 2.$$

If $\rho \approx \pm 1$:

- $Var(\hat{\beta}_1)$ & $Var(\hat{\beta}_2)$ will be large, i.e. $E\|\hat{\beta} - \beta\|^2$ will be large
- Magnitudes of $\hat{\beta}_1$ and $\hat{\beta}_2$ will be high
- Signs of $\hat{\beta}_1$ & $\hat{\beta}_2$ can be misleading.

$$\rho_{y_1} > 0$$

$$\rho_{y_2} > 0$$